

Multiple Choice

1. **Answer:** A

Options: A) BMR can be measured by the duration and quality of sleep; B) BMR can be estimated by the level of physical activity in a daily routine; C) BMR can be determined by measuring heart rate; D) BMR can be determined by the amount of sweat produced during physical activity; E) BMR can be estimated by the frequency of urination

Note: Correct option: A - BMR can be measured by the duration and quality of sleep

2. **Answer:** A

Options: A) Apical tissue; B) Sclerenchyma tissue; C) Ground tissue; D) Vascular tissue; E) Cork tissue

Note: Correct option: A - Apical tissue

3. **Answer:** C

Options: A) conjugation results in no genetic diversity; B) conjugation requires more than two organisms; C) all individuals are identical genetically, and conjugation could not occur; D) repeated fission in paramecia results in sterile offspring that cannot conjugate; E) the group of paramecia lacks the cellular structures necessary for conjugation

Note: Correct option: C - all individuals are identical genetically, and conjugation could not occur

4. **Answer:** A

Options: A) Temperature, precipitation, soil chemistry, and the types of decomposers present; B) Only soil structure and chemistry; C) Precipitation, temperature, and the presence of specific predator species; D) Only temperature and precipitation; E) Only the concentration of trace elements

Note: Correct option: A - Temperature, precipitation, soil chemistry, and the types of decomposers present

5. **Answer:** D

Options: A) Heterozygous traits; B) Mendelian traits; C) Multiple alleles traits; D) Polygenic traits; E) Pleiotropic traits

Note: Correct option: D - Polygenic traits

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Weekend hospitalization leads to delayed provision of intensive procedures and

elevated 1-year mortality for elderly AMI patients. The existence of measurable differences in treatments raises questions regarding the efficacy of a single input regulation (e.g., mandated nurse staffing ratios) in enhancing the quality of weekend care. My results suggest that targeted financial incentives might be a more cost-effective policy response than broad regulation aimed at improving quality.

7. **Answer:** False

Options: A) True; B) False

Note: GP ablation did not prove to be beneficial for postoperative stable NSR. A complete left atrial lesion set and biatrial ablation are advisable for improving rhythm outcomes. Randomized controlled trials are necessary to confirm our findings.

8. **Answer:** True

Options: A) True; B) False

Note: Interns order significantly more arterial blood gases per infant than junior and senior residents on-call in the neonatal ICU. Additional study is required to see if the experience of housestaff is associated with a broader array of neonatal outcomes, such as morbidity and mortality.

9. **Answer:** False

Options: A) True; B) False

Note: The prevalence of PAD is high in nursing home residents. AAI is not predictive for IHD mortality in this population. In very frail elderly traditional risk factors and PAD are less important predictors of death compared to poor functional status, nutritional factors and previous cardiovascular disease.

10. **Answer:** False

Options: A) True; B) False

Note: The site of access in our study does not appear to influence complications specifically neural injury or recurrence rates.

Long Form Response

11. **Answer:** causes cell death. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.89. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key